

A Highly Sensitive and Reliable Humidity Sensor for Low-Water-Pressure Detection in a Vacuum Environment via PEDOT:PSS Films

Ping Guo, Xuyang An, Xinxin He, Yinhua Hu, Jiatai Chen, Guoyun Zhou, Jinbiao Fan, Weiwei Zhang, Binbin Zhang,* Shengyuan Jiang,* and Jia Zhang*



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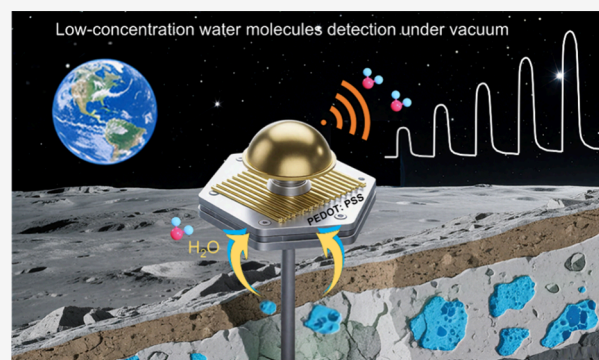
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Supporting Information

ABSTRACT: Exploring extraterrestrial water resources has far-reaching significance for future human space resource utilization and base construction. Consequently, the development of effective sensing technologies for in situ water detection is a critical endeavor. This work presents a highly sensitive and reliable water molecules sensor based on poly(3,4-ethylenedioxythiophene)/poly(styrenesulfonate) (PEDOT:PSS) films. The device demonstrates exceptional performance under high-vacuum conditions (1.0×10^{-3} Pa), achieving an extremely low detection limit of 0.02 Pa (148 $\mu\text{g/mL}$) and exhibiting outstanding reversibility, repeatability, and long-term stability. Furthermore, the sensor successfully underwent rigorous survivability tests designed to simulate the harsh conditions of space, showing no significant performance degradation. Meanwhile, through in situ film surface/thickness and spectroscopic analyses, a deeper understanding of the hydrogen-bond-based interactions between water molecules and PEDOT:PSS has been elucidated. This research provides valuable new insights and a promising platform for the development of advanced sensors aimed at detecting water on other planetary bodies.

KEYWORDS: water molecules sensor, vacuum environment, ultralow-water-pressure detection, survivability tests



Water is a basic need for life and the key to life-sustaining activities in the Earth's ecosystems.¹ In the aerospace, the exploration of the content and distribution of lunar water resources not only yields valuable insights into the lunar geological environment, but also has far-reaching significance for future lunar resource utilization and base construction.^{2–4} In 2009, the Lunar Crater Observation and Sensing Satellite (LCROSS) conducted impact experiments in the polar regions of the Moon, and the analysis results of the impact ejecta confirmed the existence of water.^{5,6} Currently, the exploration of lunar water resources mainly relies on remote sensing technology,⁷ infrared spectral analysis,^{8,9} neutron detection,¹⁰ and other means.^{11,12} However, these approaches rely on complex experimental equipment and high-energy physics detection technologies, and are plagued by limitations including insufficient detection accuracy, low spatial resolution, and a poor signal-to-noise ratio.^{13,14} Therefore, the development of miniaturized and intelligent water molecules sensors adapted to the Moon has become an urgent need for lunar resource utilization studies.^{15–17} If realized, such sensors could be mounted on lunar rovers or fixed scientific payloads to monitor the process of heating and releasing water from the lunar soil in real time. Combined with orbital data, this would enable the construction of high-precision water distribution models.

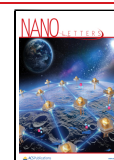
Due to the special characteristics of the lunar environment, the technical requirements for the applied sensors are more stringent.^{18–20} The lunar surface is in a near-absolute vacuum, necessitating sensors to operate reliably in such an environment. Additionally, the extremely low content of extraterrestrial water molecules demands water molecules sensors with a low detection limit, high sensitivity, high stability and repeatability. More importantly, sensors are subjected to intense shocks and vibrations during launch, landing, and contact with extraterrestrial surfaces, and may be exposed to extreme conditions (e.g., prolonged high humidity and temperature fluctuations) in space.^{21,22} These factors may lead to sensor deactivation or performance degradation. Therefore, to ensure the reliable operation of sensors under harsh extraterrestrial conditions, precise design and rigorous testing are imperative to guarantee reliability and accuracy across diverse environments.

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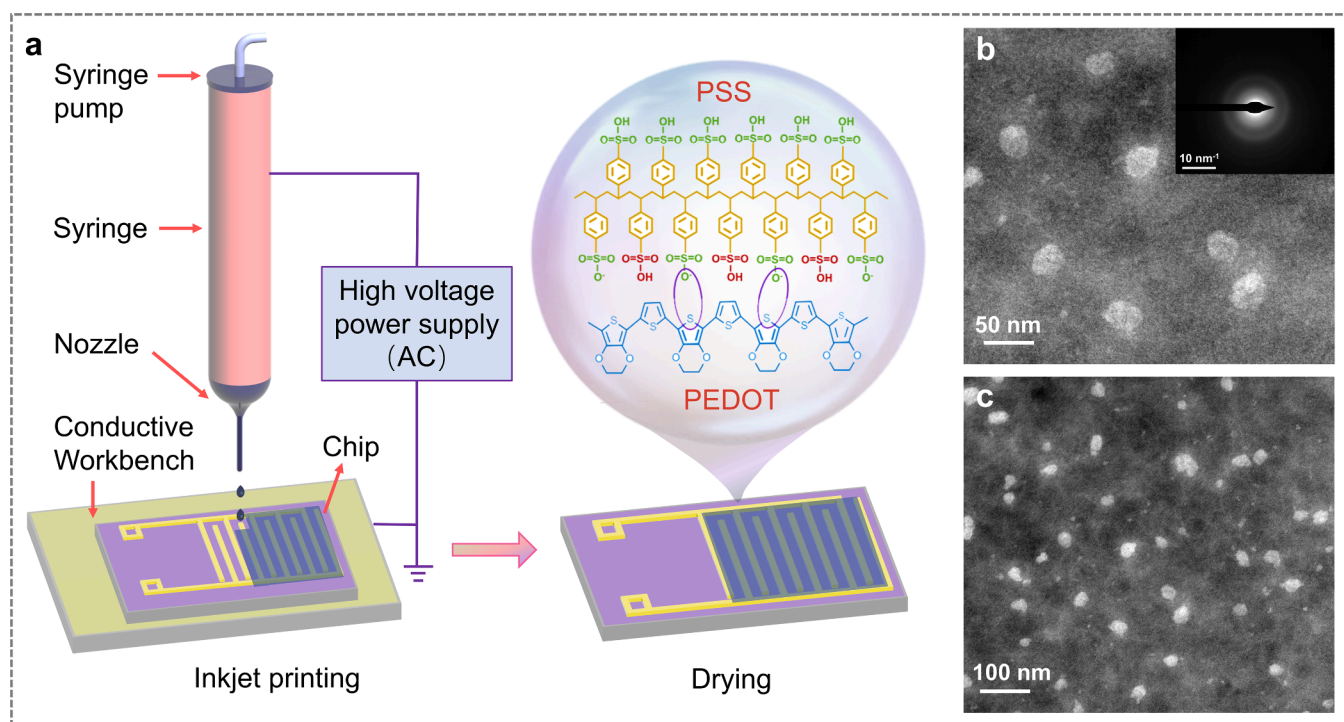


Figure 1. Schematic diagram of manufacturing process and morphological characterization. (a) Schematic diagram of the PEDOT:PSS sensor manufacturing process and the molecular structure of PEDOT:PSS. (b and c) HAADF-STEM images of a PEDOT:PSS film with different magnifications.

Poly(3,4-ethylenedioxythiophene)/poly(styrenesulfonate) (PEDOT:PSS), a common conductive polymer, has gained growing attention as a sensitive material for water molecules sensors.^{23,24} Owing to the hydrophilicity of PSS, slight ambient humidity variations induce significant changes to the conductivity of PEDOT:PSS,^{25,26} endowing it with high sensitivity and responsiveness in water molecules sensors.^{27,28} Additionally, as a commercially available material with excellent conductivity, good mechanical flexibility, and strong environmental stability,²⁹ PEDOT:PSS holds great potential for aerospace applications. However, no research has explored PEDOT:PSS sensors for detecting extraterrestrial trace water molecules in the space vacuum environment.

Herein, we report a highly sensitive and reliable water molecules sensor based on PEDOT:PSS films. The sensor is capable of detecting water molecules under high vacuum (1.0×10^{-3} Pa). The PEDOT:PSS sensor exhibits an extremely low lower detection limit of 0.02 Pa (equal to 148 $\mu\text{g/mL}$), along with excellent detection reversibility, repeatability, and long-term stability. In addition, in order to verify that the sensor has the potential to function properly under aerospace conditions, it was subjected to rigorous survivability tests, and showed no significant performance degradation or deactivation after the tests. This work provides new insights into the development of effective sensors for the detection of extraterrestrial water resources.

As a well-known conductive polymer, PEDOT:PSS features a core-shell grain structure. As illustrated in Figure 1a, its PEDOT main chain is composed of interconnected thiophene rings, forming a continuous conjugated system through π - π bonds.³⁰ PSS, a polymer derived from free-radical polymerization of styrene monomers, acts as a charge stabilizer.^{31,32} Due to the hydrophilic nature of PSS, the electrical resistance of PEDOT:PSS films is susceptible to ambient humidity. The

absorption of water molecules may swell the PSS shells, resulting in a larger distance between neighboring PEDOT-enriched cores, which increases the resistance.^{33,34} Based on the above principle, PEDOT:PSS was selected as the sensitive material for water molecules detection in vacuum environments. To fabricate the water molecules sensor, an electrohydrodynamic jet printing technique was used to deposit the PEDOT:PSS dispersion onto the electrode surface, with a 60 μm dispensing needle employed during printing to ensure precise and uniform ink deposition. Figure S1 shows that the sensitive material was uniformly deposited on the interdigital electrode, with clear boundaries and no surface defects. Besides, transmission electron microscopy (TEM) imaging of PEDOT:PSS was carried out at room temperature. Figures S2 and 1b,c reveal the bright-field TEM (BF-TEM) images and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images (with better contrast) of PEDOT:PSS, and it can be clearly observed that individual grains have a near-shell-core structure, with an average diameter of 30–50 nm, which is in agreement with previous reports in the literature.³⁵

To evaluate the performance of the PEDOT:PSS sensor under vacuum conditions, a simulated lunar detection system (Figure S3) was designed, comprising a detection chamber, water supply pipelines, a source introduction system [controlled via an atomic layer deposition (ALD) valve], a water source control box, a baffle valve, a mechanical pump, a molecular pump, a compound vacuum gauge, and a source meter. Inside the chamber, a sample stage holds the sensor, a pipeline above the sample stage connects to the water source introduction system, and a nitrogen bottle is connected to an ALD pneumatic valve. Water molecules with different concentrations are supplied via the water control box. The chamber's bottom connects to a molecular pump (fitted with a

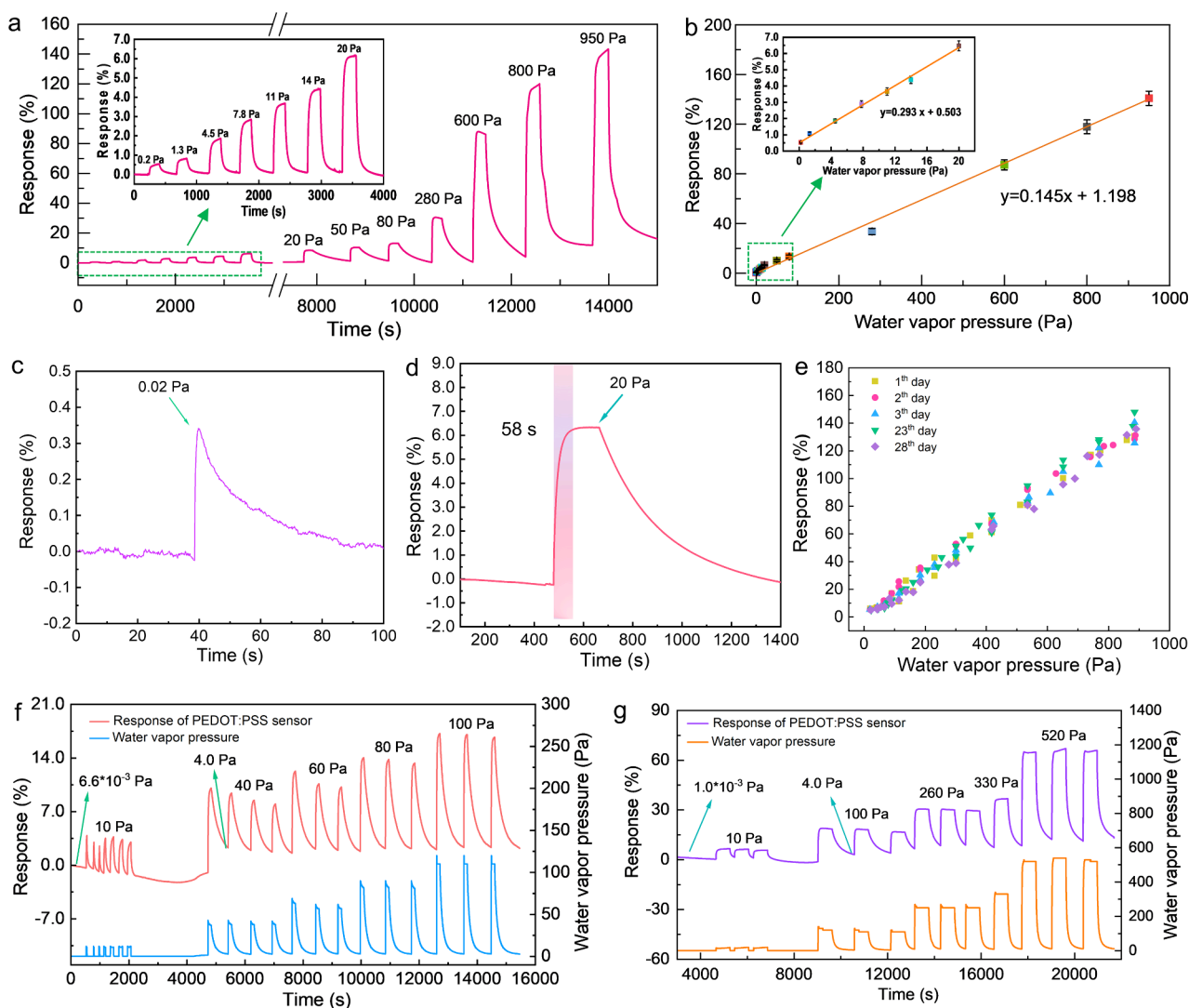


Figure 2. Sensing performance of the PEDOT:PSS water molecules sensor under a vacuum of 1.0×10^{-3} Pa at an operating temperature of 293.15 K. (a) Dynamic response–recovery curve over the range of 0.2–950 Pa. (b) Linear fit of the sensor's response with water molecules pressure. (c) Response curve of the sensor under a water molecules pressure of 0.02 Pa. (d) Response and recovery time of the sensor under a water molecules pressure of 20 Pa. (e) Sensor's response as a function of water molecules pressure over a 1-month period. (f) Dynamic response–recovery curve of the sensor to repeated exposures to water molecules at 10, 40, 60, 80, and 100 Pa, holding for 60 s each time. (g) Dynamic response–recovery curve of the sensor to repeated exposures to water molecules at 10, 100, 260, 330, and 520 Pa, holding for 600 s each time.

front-end baffle valve to control pressure maintenance during pulsed water introduction and vacuum restoration in the desorption phase) and a mechanical pump, with a compound vacuum gauge monitoring the chamber's real-time vacuum level. The device enables controlled water vapor transport from 0.01 to 1000 Pa under a vacuum of 1×10^{-3} Pa, with the transport resolution as low as 0.01 Pa. The system is also equipped with a temperature control range of -80 to -250 °C and a laser calibration inlet and outlet (1292.5 nm diode laser) to accurately determine water molecules concentrations in the chamber.

As shown in Figure 2a, the PEDOT:PSS sensor was tested for dynamic response to water molecules across a pressure range of 0.2–950 Pa (1.48 mg/mL to 7.03 g/mL) under a vacuum of 1.0×10^{-3} Pa. Results indicate that the sensor's response increases from 0.53% to 141% with rising water vapor pressure, without reaching saturation. Furthermore, the sensor exhibits baseline resistance recovery of $\geq 99.7\%$ for the 0.2–20 Pa range and $\geq 93.43\%$ for the 20–950 Pa range after each

measurement cycle, further confirming its excellent recoverability and reliability. Figure 2b illustrates a strong linear relationship between the response and water molecules pressure in each range, with 20 Pa as the cutoff. The sensitivity is 0.290%/Pa for pressures below 20 Pa and 0.146%/Pa for 20–950 Pa, indicating higher sensitivity at low water molecules pressures. Figure 2c shows the sensor's response curve for 0.02 Pa (148 $\mu\text{g/mL}$) water molecules with a response of approximately 0.33%, demonstrating its ability to detect extremely low water molecules pressures. The detection limit (<0.02 Pa) is one-fifth of the lowest reported detection level to date (Table S1). To evaluate the response time of the sensor, the response curve in Figure 2d was plotted for 20 Pa water molecules. It takes 58 s from the introduction of the water molecules to the saturation of the sensor's response, which indicates a fast response time. Besides, the long-term stability of the sensor was assessed by measuring its response changes to 10–900 Pa water molecules over one month. The results (Figures 2e and S4) demonstrate consistent response levels

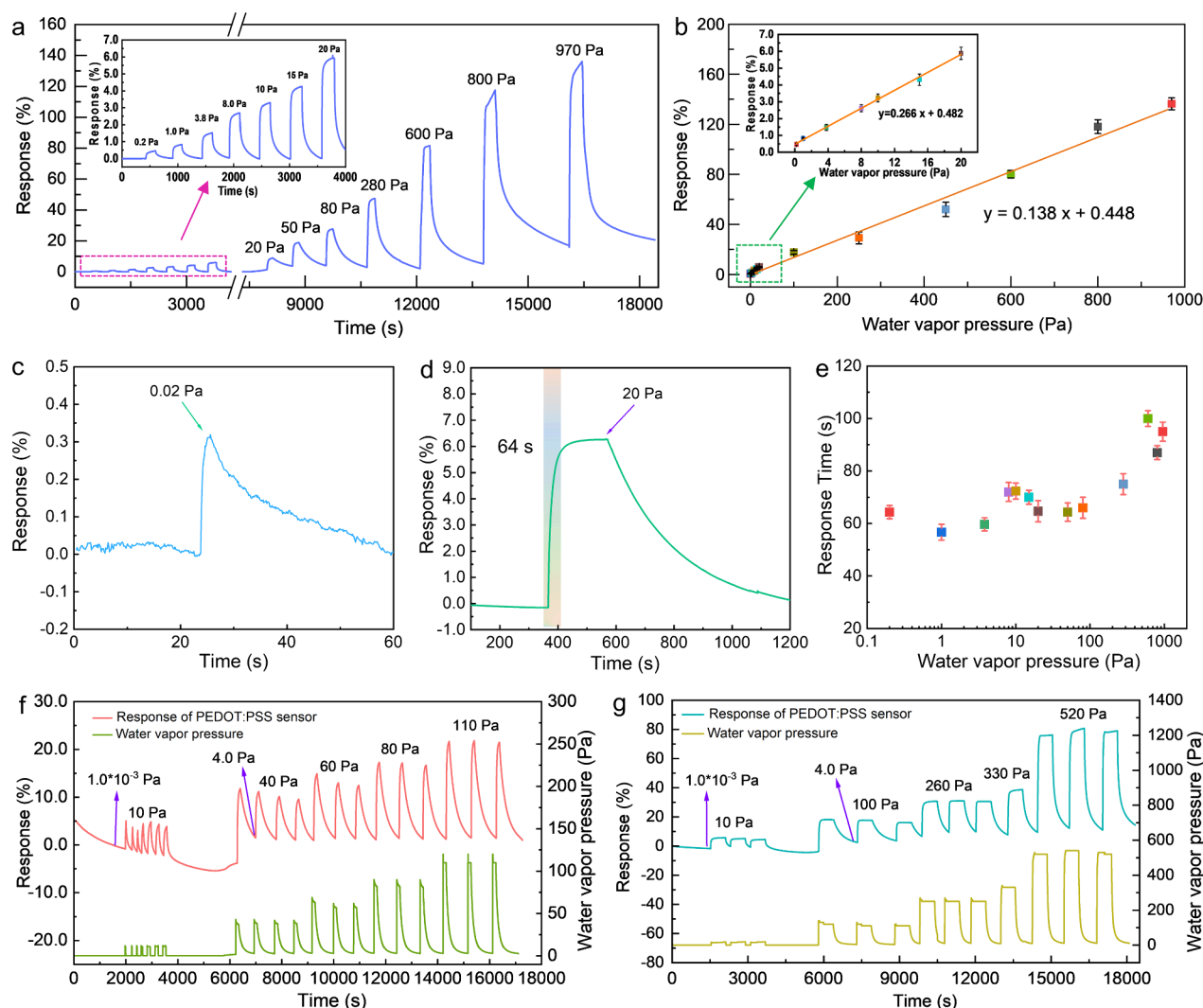


Figure 3. Sensing performance of the PEDOT:PSS water molecules sensor under a vacuum of 1.0×10^{-3} Pa at an operating temperature of 293.15 K after survivability tests. (a) Dynamic response–recovery curve over the range of 0.2–950 Pa. (b) Linear fit of the sensor's response with water molecules pressure. (c) Response curve of the sensor under a water molecules pressure of 0.02 Pa. (d) Response time of the sensor under a water molecules pressure of 20 Pa. (e) Variation of the sensor response time with water molecules pressure. (f) Dynamic response–recovery curve of the sensor to repeated exposures to water molecules at 10, 40, 60, 80, and 100 Pa, holding for 60 s each time. (g) Dynamic response–recovery curves of the sensor to repeated exposures to water molecules at 10, 100, 260, 330, and 520 Pa, holding for 600 s each time.

with a difference of 1.59%, confirming its excellent long-term stability. The dynamic response of the sensor is shown in Figure 2f for repeated exposures to water molecules at 10, 40, 60, 80, and 100 Pa, each with a holding time of 60 s. The sensor's consistent response to the same pressure of water molecules with no drift in the baseline is evidence of the device's excellent repeatability and consistency. For further repeatability evaluation, the sensor was exposed to water molecules for 600 s (Figure 2g), showing maximum amplitude variations of 4.74%, 3.76%, and 2.84% for repeated 10, 260, and 520 Pa water vapor exposures, respectively. Besides, the sensor demonstrates minimal detection hysteresis and high detection accuracy in vacuum environments, offering significant advantages for practical applications (Figure S5). Overall, all tests collectively demonstrate that the PEDOT:PSS sensor features a wide detection range, extremely low detection limit, excellent reversibility, repeatability, and long-term stability, making it a highly reliable tool for water molecules detection in vacuum environments. Additionally, Table S1

shows the comparison of previously published PEDOT:PSS-based sensors for humidity detection, further verifying that the sensor in this work exhibits exceptional water molecules sensing capability under vacuum.

To validate the feasibility of the PEDOT:PSS sensor for future aerospace conditions detection, the sensor underwent rigorous survival tests, including impact testing, atmospheric/vacuum high/low temperature cycling, and humidity resistance testing (Figures S6–S8). After these tests, the sensor's performance was re-evaluated and compared to its pretest performance to assess its stability and reliability. Figure 3a shows the dynamic response of the sensor to water molecules in the range of 0.2–950 Pa (1.48 mg/mL to 7.03 g/mL). The sensor's response still increases with water molecules pressures without reaching saturation and exhibits excellent recoverability after the survival tests. The sensitivity is 0.266%/Pa for water molecules pressures below 20 Pa and 0.137%/Pa for pressures between 20 and 950 Pa (Figure 3b). Although the sensor's sensitivity in each partial pressure range is slightly

lower than the presurvivability test levels, its response and linearity remain excellent. Figure 3c demonstrates the sensor's response curve for 0.02 Pa (148 $\mu\text{g/mL}$) water molecules with a response of approximately 0.28%, indicating that its detection limit remains below 0.02 Pa after the survival tests. Parts d and e of Figure 3 plot the response times of the sensor exposed to different water molecules pressures. Response times range from 55 to 70 s for pressures below 280 Pa and increase to nearly 100 s at higher pressures. Additionally, the sensor's stability was retested under short-term (60 s) and long-term (600 s) holding at different water molecules pressures (Figure 3f,g). The sensor shows consistent responses at the same pressures (no baseline drift) and highly repeatable results. These performance tests collectively indicate that the PEDOT:PSS sensor exhibits no significant performance degradation or deactivation after the rigorous survival tests, supporting its potential for future aerospace condition detection. In addition, the sensing performance tests at both a low temperature of 2 $^{\circ}\text{C}$ (Figure S9) and a high temperature of 100 $^{\circ}\text{C}$ (Figure S10) were conducted to further evaluate the ability of the sensor to operate under vacuum. Although the sensor exhibits temperature-dependent sensitivity variations, it maintains excellent linear response and repeatability characteristics. This predictable behavior allows for effective temperature compensation through a precalibrated sensitivity–temperature relationship.

A detailed explanation of the mechanism for water molecules detection is provided in Figure 4a. Typically, every three to

dominated by Regime 1 in Figure 4b, and the following process (eq 1) occurs:



As the hydrogen bonds between the PSS chains are weakened, the volume of the PSS expands, leading to an increase in the distance between the PEDOT cores (the conductive part of the polymer) as well.^{38,39} It becomes more difficult for carriers to move within the polymer, thus increasing the resistance of the film. When the water molecules are separated from the PSS, the volume of the PSS returns to its initial state, the distance between the PEDOT chains recovers, so the resistance returns to its initial value. Based on the above principle, PEDOT:PSS undergoes reversible changes in volume and conductivity upon interaction with water molecules, enabling its excellent sensing performance under vacuum.

To verify the above sensing mechanism and gain deeper insight into the interaction between water molecules and PEDOT:PSS, in situ film thickness analyses and in situ spectroscopic analyses were performed. First, white-light interferometry (WLI) measurements were carried out on the dried film and the film after water vapor exposure. A 150 $\mu\text{m} \times 150 \mu\text{m}$ area in the center of the film was selected for scanning. For the initially dried film, a 2D image of its surface profile is shown in Figure 5a, with an average surface roughness of 37 nm over a selected length range (Figure 5c). Subsequently, a humidifier was used to continuously humidify the area around the film, and the surface profile at the same location that was rescanned with WLI, is shown in Figure 5b, where the average surface roughness increases to 52.6 nm (Figure 5c). The above results indirectly prove that PEDOT:PSS films interact with water molecules and undergo hygroscopic expansion, leading to an increase in surface roughness. Then, in order to quantitatively measure the absolute thickness change, the film thickness under different humidity conditions was measured using a stylus profilometer. As shown in Figure 5f, the initially dried film surface (average thickness: 510.4 nm) undergoes hygroscopic swelling with increasing ambient humidity, exhibiting average thicknesses of 516.4 nm (37% RH), 539.3 nm (54% RH), 554.5 nm (71% RH), and 584.4 nm (89% RH), respectively. The above experimental results are consistent with the mechanism explanation part. In addition, to verify the reversibility of the film's hygroscopic change, the thickness of the PEDOT:PSS film was remeasured after half an hour drying, and the result showed that the film thickness recovered to 501.5 nm (Figure 5i). This confirms that PSS reverts to its initial volume when water molecules detach, supporting the sensor's reversible sensing performance under vacuum.

Fourier transform infrared (FTIR) spectra of the PEDOT:PSS film in dry and saturated water molecules environments were tested to reveal distinct vibrational signatures, which elucidate the hydration mechanism (Figure 5d). In the dry state, the FTIR spectra exhibit characteristic absorption bands corresponding to PEDOT and PSS components, including C=C conjugation vibrations of the thiophene ring at 1529 cm^{-1} , C–O stretching at 1265 cm^{-1} ,^{40,41} and S=O symmetry vibrations of the SO_3H and SO_3^- groups in PSS at 1009.5 and 1036 cm^{-1} .^{42,43} When the PEDOT:PSS film was exposed to saturated water vapor environment, in addition to the absorption bands from PEDOT and PSS, a new peak corresponding to the bending

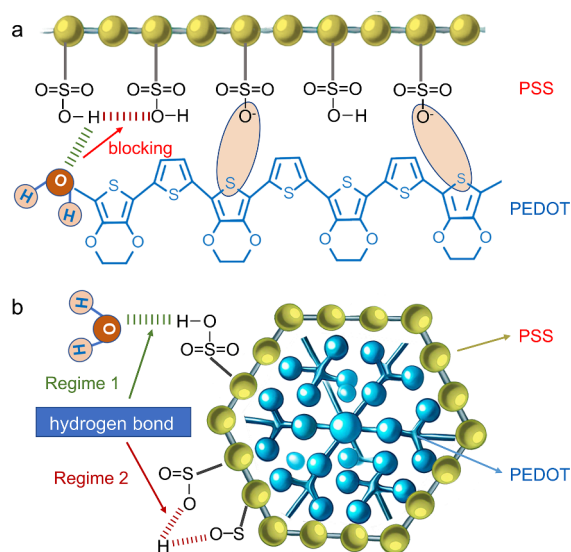


Figure 4. Schematic of the PEDOT:PSS water molecules sensor detection mechanism. (a) Schematic diagram of the molecular structure of PEDOT:PSS and its interaction with water molecules. (b) Microstructure changes of PSS chains upon combination with water molecules.

four thiophene units in the PEDOT chain carry a positive charge, while the benzene ring in the PSS chain has two sulfonic acid groups that can ionize into SO_3^- or neutral SO_3H .^{27,36} The strong cohesion between PSS chains stems from intermolecular or intramolecular hydrogen bonding within the chains. When water molecules are present, they will hydrate with free SO_3H through hydrogen bonding. At room temperature, each PSS chain will absorb three or four water molecules, inhibiting the formation of hydrogen bonds between SO_3H groups in PSS.^{26,37} The microstructure will be

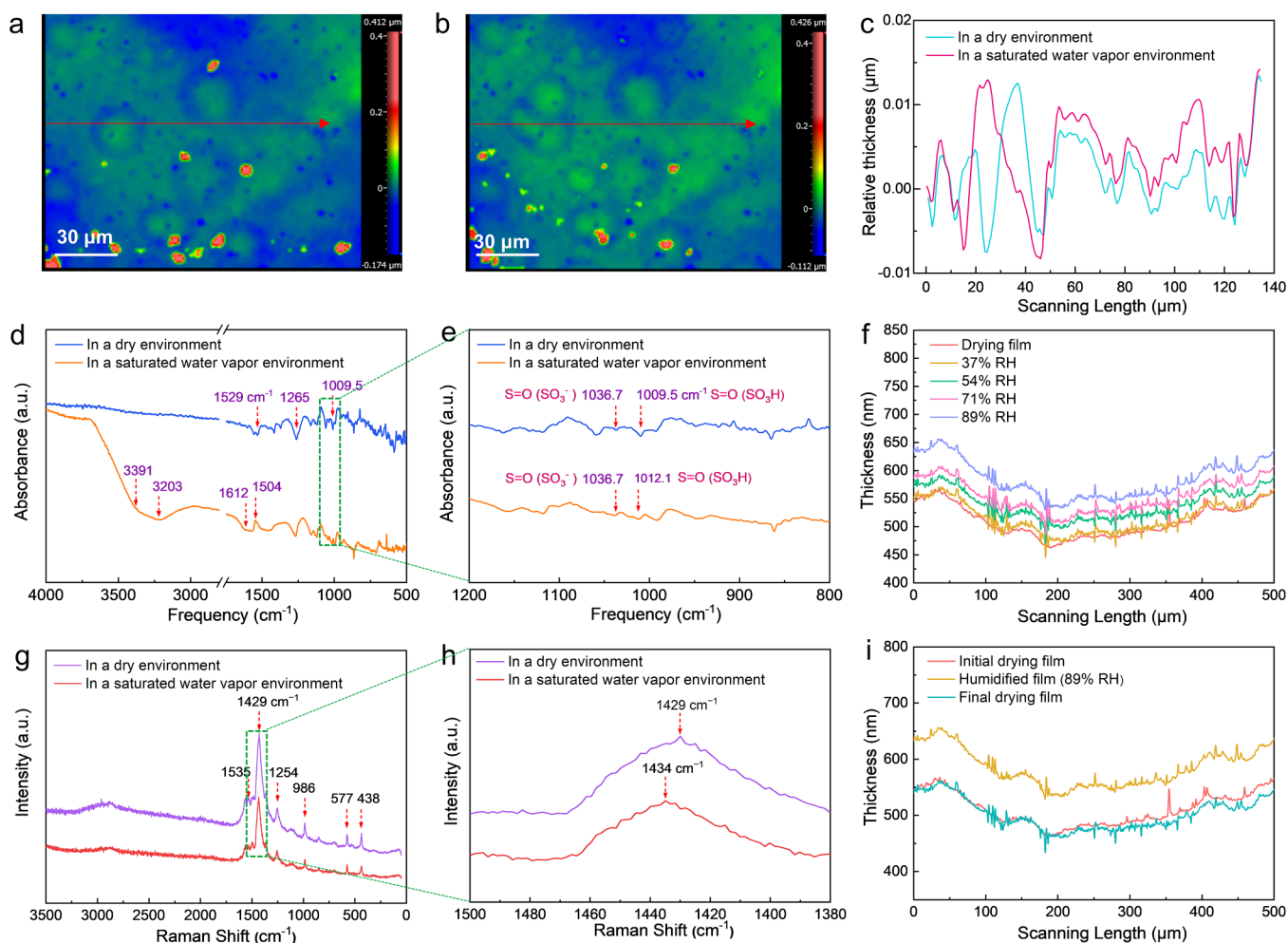


Figure 5. Characterization of the PEDOT:PSS water molecules sensor detection mechanism. WLI measurements on dry PEDOT:PSS films (a) before and (b) after water vapor exposure. (c) Relative thickness variation over the selected length range (red lines in parts a and b). (d) FTIR spectra. (e) Enlarged view of the FTIR spectra of the PEDOT:PSS film in dry conditions and in a saturated water vapor environment. (f) Thickness changes of the PEDOT:PSS film measured using a profilometer under different humidity conditions. (g) Raman spectra. (h) Enlarged view of the Raman spectra of the PEDOT:PSS film in dry conditions and in a saturated water vapor environment. (i) Thickness changes of the PEDOT:PSS film measured using a profilometer after drying resumption.

vibration of H_3O^{+44} was observed at 1600 cm^{-1} . More importantly, when water molecules are adsorbed, a strong $\text{C}=\text{C}$ vibrational absorption band in the opposite direction is observed at $1550\text{--}1510\text{ cm}^{-1}$, which is associated with the formation of bipoles, a key feature of the electrical conductivity of PEDOT:PSS materials.⁴⁵ In addition, the enlarged view of this FTIR spectrum offers deeper insight into the interaction of the sulfonic acid groups in PSS with water based on hydrogen bonding. As shown in Figure 5e, the intensity of the $\text{S}=\text{O}$ bond peak in HSO_3^- is weakened and blue-shifted when there is adsorption of water molecules, which proves that the water molecules undergo proton transfer with SO_3H (eq 1).⁴⁶ Specifically, during the process of proton transfer involving water, the proportion of SO_3^- increases and the proportion of SO_3H decreases, and because the symmetrical vibrational frequency of SO_3^- ($\sim 1036\text{ cm}^{-1}$) is higher than that of SO_3H , it is manifested by a weakening of the intensity of the peaks of $\text{S}=\text{O}$ and a shift of the peaks toward higher wavenumbers in HSO_3^- .^{44,45} Finally, Raman spectroscopy was used to assist in verifying the above mechanisms. In the dry environment, the symmetric and asymmetric $\text{C}=\text{C}$ vibrations of the thiophene ring in PEDOT corresponded to peaks near 1429 and 1535

cm^{-1} , respectively.^{47–49} (Figure 5g). For PEDOT:PSS films in a saturated water molecules environment, a shift of the $\text{C}=\text{C}$ symmetric vibration at 1429 cm^{-1} toward higher wavenumbers to 1434 cm^{-1} can be observed (Figure 5h). This originates from the hydrogen-bonding interaction with the $-\text{SO}_3\text{H}$ in PSS after water adsorption, which causes the partial dedoping of PEDOT and weakens the electrostatic interactions between the PSS and PEDOT chains.^{35,50} Overall, the above in situ analyses clarify the interaction between PEDOT:PSS and water molecules, confirming its ability to be used for the detection of water molecules under vacuum.

In summary, a highly sensitive and reliable sensor based on PEDOT:PSS for trace water molecules detection was developed. The sensor is capable of detecting water molecules under high vacuum ($1.0 \times 10^{-3}\text{ Pa}$). It demonstrates an exceptionally low detection limit of 0.02 Pa ($148\text{ }\mu\text{g/mL}$) and outstanding detection reversibility, repeatability, and long-term stability. Furthermore, to evaluate its performance under aerospace conditions, the sensor underwent rigorous survivability tests, showing no significant degradation in performance or deactivation after testing. Meanwhile, through in situ film surface/thickness and spectroscopic analyses, a deeper under-

standing of the hydrogen-bond-based interactions between water molecules and PEDOT:PSS has been gained. This work represents a significant step forward in the development of advanced sensors that can support the exploration and utilization of extraterrestrial water resources, paving the way for future space exploration and base construction.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c03201>.

Details of experimental materials, gas sensor fabrication and measurement methods, the survivability test process, optical microscopy and BF-TEM images, schematic diagram of a vacuum environment detection system, response on the 1st and 28th days as a function of water pressure, individual thermal cycle procedures in survival tests, dynamic response–recovery curves, and repeatability tests at 2 and 100 °C (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jia Zhang — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China; State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150080, China; Chongqing Research Institute, Harbin Institute of Technology, Chongqing 401120, China; orcid.org/0000-0003-0943-7543; Email: zhangjia@hit.edu.cn

Shengyuan Jiang — State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150080, China; Email: jiangshy@hit.edu.cn

Binbin Zhang — School of Materials and Energy, University of Electronic Science and Technology of China, Chengdu 610054, China; Email: zhang0001234@163.com

Authors

Ping Guo — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China

Xuyang An — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China

Xinxin He — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China

Yinhua Hu — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China

Jiatai Chen — Key Laboratory of Microsystems and Microstructure Manufacturing, Ministry of Education, Harbin Institute of Technology, Harbin 150080, China

Guoyun Zhou — School of Materials and Energy, University of Electronic Science and Technology of China, Chengdu 610054, China; orcid.org/0009-0006-3229-7912

Jinbiao Fan — Science and Technology on Electronic Test and Measurement Laboratory, North University of China, Taiyuan 030051, China

Weiwei Zhang — State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150080, China

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.nanolett.5c03201>

Author Contributions

The project was hosted by J.Z., B.Z., and S.J.. P.G. designed and conducted the experiments and analyzed the data. X.A., X.H., Y.H., and J.C. did some of measurements and analysis. W.Z., G.Z., and J.F. rewrote the manuscript and gave constructive suggestions. All authors analyzed and discussed the results and contributed to the final manuscript.

Notes

The authors declare no competing financial interest.

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